

Vishwa Nuwan

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SUMMARY

Final-year Software Engineering undergraduate at SLTC with hands-on experience building enterprise-grade full-stack applications, distributed systems, and AI-powered platforms. Driven by solving complex engineering problems and focused on designing scalable software solutions and delivering end-to-end products across web, backend, cloud, and machine learning domains.

EDUCATION

Bachelor of Science (Honours) in Software Engineering

Expected November 2026

Sri Lanka Technology Campus (SLTC)

Relevant Coursework: Software Engineering, Web Technologies, Database Systems, Machine Learning, Cloud Computing, Data Structures and Algorithms

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, C++, C#, PHP, HTML, CSS, SQL

Frontend: React.js, Tailwind CSS, Bootstrap, Redux Toolkit, Vite

Backend & APIs: Node.js, Express.js, Spring Boot, ASP.NET Core, FastAPI, REST APIs, AI APIs, SignalR

Databases: MySQL, PostgreSQL, SQLite, MongoDB, Firebase, Prisma ORM, Entity Framework Core

ML & Data: scikit-learn, MLflow, pandas, NumPy, XGBoost, Apache Airflow, DVC

Cloud & DevOps: AWS, Docker, Kubernetes, RabbitMQ, GitHub Actions, Git, Postman, JWT, Cloudinary

PROJECTS AND EXPERIENCE

Vision POS - POS & ERP System (React · Java · Redis · RabbitMQ · MySQL · Kubernetes)

2026 - Present

- Developed a production-grade multi-tenant POS for the Sri Lankan retail market across 60+ REST endpoints, 50+ JPA entities, and 11 Flyway migrations, supporting VAT/NBT/SSCL compliance, full E-Invoice lifecycle, trilingual UI (EN/SI/TA) via i18next, and local payment gateway integrations (Mobitel mCash, EzCash, Frimi)
- Engineered a Java backend using RabbitMQ for receipt, audit, and report workflows, WebSocket/STOMP for KDS notifications, and Redis for JWT blacklisting, rate limiting, and TTL session caching, alongside a Flutter Cashier/KDS app with BCrypt security, RBAC, AOP auditing, and CI security scanning (OWASP, Trivy, Semgrep)

VisionWMS - Enterprise WMS (Java · React · PostgreSQL · Redis · RabbitMQ · Docker)

2026 - Present

- Engineered a full-stack enterprise WMS spanning 42+ business modules across the complete warehouse lifecycle, backed by 35 Flyway migrations, featuring wave-optimized picking, FIFO/FEFO batch allocation, Redis-based stock reservations, real-time shipment tracking, and scheduled background jobs for inventory alerts, auto-replenishment, and demand forecasting
- Architected a Java-based backend with an event-driven architecture using RabbitMQ for replenishment, ASN, email, and stock movement workflows, WebSocket/STOMP for real-time tracking and notifications, JWT dual-token auth (1h access, 7d refresh) with Redis blacklisting, BCrypt (12) security, RBAC (4 roles), Bucket4j rate limiting, and JPA auditing with soft-delete across 50+ entities

AI-Powered Code Reviewer Tool (Java · C++ · Docker · RabbitMQ · React · PostgreSQL)

2026

- Built a microservices-based GitHub PR analysis platform across 14 containerized services (7 infrastructure, 7 application), orchestrated via Docker Compose with 3 isolated PostgreSQL instances, RabbitMQ, and Redis
- Engineered a Java Spring Boot AST parser with BFS/DFS traversal, feeding structured PR context to Claude AI for code reviews, alongside a containerized C++ static analyzer for memory leak detection, integrated via an event-driven RabbitMQ pipeline

Numo - Real-Time Chat App (React · ASP.NET Core · SignalR · RabbitMQ · Docker)

2026

- Architected a microservices-based chat application across 4 independently deployable services supporting DMs, group channels, voice/video calls, AI chat, and media sharing, using JWT authentication, Entity Framework Core for data persistence, Docker containerization, and a Java-based media service for scalable file handling, integrated via RabbitMQ and SignalR for real-time event-driven communication services

Royal - Hostel Management System (PHP 8.0 · MySQL · Bootstrap 5 · jQuery)

2026

- Developed a full-stack role-based hostel management platform with admin and tenant dashboards for booking, occupancy tracking, and maintenance, backed by a normalized MySQL schema with 16 database constraints
- Engineered a security-focused backend with 23 parameterized SQL queries, BCrypt password hashing, token-based password resets (24-hour expiry), race-condition-safe bookings using SELECT FOR UPDATE, and automated email notifications via PHPMailer